

## MODULE 3: SCIENTIFIC MANAGEMENT - TAYLOR'S CONTRIBUTION

### F.W. TAYLOR'S CONTRIBUTION TO MANAGEMENT

#### Introduction to Taylor and his work

*Fredrick Winslow Taylor (March 20, 1856 - March 21, 1915)* commonly known as “**Father of Scientific Management**” started his career as an operator and rose to the position of chief engineer. He conducted various experiments during this process which forms the basis of scientific management. It implies application of scientific principles for studying & identifying management problems. He sought to create a **mental revolution** among both workers and managers by defining clear guidelines

According to Taylor, “Scientific Management is an art of knowing exactly what you want your men to do and seeing that they do it in the best and cheapest way”. In Taylor's view, if a work is analyzed scientifically it will be possible to find *one best way* to do it.

- Taylor's scientific management is popularly called as “taylorism”
- Scientific management involves the application of a scientific approach to management decision-making; and discarding at the same time, all unscientific approach as like rule of the thumb, a hit or miss approach and a trial and error approach.

According to *Drucker*, “The cost of scientific management is the organized study of work, the analysis of work into simplest element & systematic management of worker's performance of each element”.

#### PRINCIPLES OF SCIENTIFIC MANAGEMENT:

##### 1. Science, not the rule of thumb

The basic principle of scientific management is the adoption of a scientific approach to managerial decision making and a complete discard of all unscientific approaches.

##### 2. Harmony, not discord

Harmony refers to the unity of action, while discords refer to differences in approach.

### **3. Co-operation, not individualism**

Co-operation refers to working, on the part of people towards the attainment of group objectives, while regarding their individual objectives as subordinate to the general interest.

### **4. Maximum production, in place of restricted production**

His view the most dangerous evil of the industrial system was a deliberate restriction of output.

### **5. Development of each person to the greatest of his capabilities**

Management must endeavour to develop people to the greatest of their capabilities to ensure maximum prosperity for both employees and employers.

### **6. A more equal division of responsibility between management and workers**

This principle of scientific management recommends a separation of planning from executives.

### **7. Mental revolution on the part of managerial and workers**

It involves a complete mental revolution on the part of both sides to industry viz. workers and management

## **FEATURES OF SCIENTIFIC MANAGEMENT**

- 1. Separation of Planning and Doing.** Taylor emphasised the separation of planning aspect from actual doing the work. Before Taylor's scientific management, a worker used to plan about how he had to work and what instruments were necessary for that. Taylor emphasised that planning should be left to supervisor and worker should emphasise only operational work.
- 2. Functional foremanship.** Taylor evolved the concept of functional foremanship based on specialization of functions. In this system, eight persons are involved to direct the activities of worker. Out of these, four persons are concerned with planning (i) route clerk, (ii) instruction card clerk, (iii) time and cost clerk, (iv) disciplinarian. The remaining four persons are concerned with doing aspect of the work. These are: (i) speed boss, (ii) inspector, (iii) maintenance foremen, and (iv) gang boss. All of them give directions to workers on different aspects of work.
- 3. Job analysis.** Job analysis is undertaken to find out the one best way of doing the thing. The best way of doing a job is one which requires the least movements, consequently less time

and cost. The best way of doing the thing can be determined by taking up time- motion- fatigue studies.

4. **Standardization.** As far as possible, standardization should be maintained in respect of instruments and tools, period of work, amount of work, working conditions, cost of production, etc.
5. **Scientific Selection and Training of Workers.** Taylor has suggested that workers should be selected on scientific basis taking into account their education, work experience, aptitude, physical strength, etc.
6. **Financial Incentives.** Financial incentives can motivate workers to put in their maximum efforts. If provisions exist to earn higher wages by putting in extra effort, workers will be motivated to earn more. Taylor himself applied the concept of differential piece rate system which was highly motivating.
7. **Economy.** While applying scientific management, not only scientific and technical aspects should be considered but adequate consideration should be given to economy and profit. For this purpose, techniques of cost estimates and control should be adopted. The economy and profit can be achieved by making the resources more productive as well as by eliminating the wastages.
8. **Mental Revolution.** Scientific management depends on the mutual co-operation between management and workers. For this co-operation, there should be mental change in both parties from conflict to co-operation. Taylor feels that this is the most important feature of scientific management because in its absence, no principle of scientific management can be applied.

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### **Caselet1: Scientific Management**

'F' limited was engaged in the business of food processing and selling its products under a popular brand. Lately the business was expanding due to good quality and reasonable prices. Also with more people working the market for processed food was increasing. New players were also coming to cash in on the new trend. In order to keep its market share in the short run the company directed its existing workforce to work overtime. But this resulted in many problems. Due to increased pressure of work the efficiency of the workers declined. Sometimes the subordinates had to work for more than one superior resulting in declining efficiency. The divisions that were previously working on one product were also made to work on two or more products. This resulted in a lot of overlapping and wastage. The workers were becoming undisciplined. The spirit of teamwork, which had characterised the company, previously was beginning to wane. Workers were feeling cheated and initiative was declining. The quality of the products was beginning to decline and market share was on the verge of decrease. Actually the company had implemented changes without creating the required infrastructure.

The management of company 'F' Limited now realised its folly. In order to rectify the situation it appointed a management consultant 'M' consultants to recommend a restructure plan to bring the company back on the rails. 'M' consultants undertook a study of the production process at the plant of the company 'F' limited and recommended the following changes:

- a) The company should introduce scientific management with regard to production.
- b) Production Planning including routing, scheduling, dispatching and feedback should be implemented.
- c) In order to separate planning from operational management 'Functional foremanship' should be introduced.
- d) 'Work study' should be undertaken to optimise the use of resources.
- e) 'Standardization' of all activities should be implemented to increase efficiency and accountability.
- f) To motivate the workers 'Differential Piece Rate System' should be implemented.

It was expected that the changes will bring about a radical transformation in the working of the company and it will regain its pristine glory.

### **Questions**

1. Do you think that introduction of scientific management as recommended by M consultants will result in intended outcome?
2. What precautions should the company undertake to implement the changes?

3. Give your answer with regard to each technique separately as enunciated in points a through f in the case problem.